AWS , EKS & kubectl Installed

```
abi@ubuntu:~$ sudo su -
[sudo] password for abi:
root@ubuntu:~# aws --version
aws-cli/2.33.12 Python/3.13.11 Linux/6.8.0-101-generic exe/x86_64.ubuntu.24
root@ubuntu:~# curl --location "https://github.com/weaveworks/eksctl/releases/latest/download/eksctl_Linux_amd64.tar.gz" -o eksctl.tar.gz

tar -xzf eksctl.tar.gz
sudo mv eksctl /usr/local/bin
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0
  0     0    0     0    0     0      0      0 --:--:--  0:00:01 --:--:--     0
  0     0    0     0    0     0      0      0 --:--:--  0:00:01 --:--:--     0
100 35.4M  100 35.4M    0     0   389k      0  0:01:33  0:01:33 --:--:-- 4373k
root@ubuntu:~# eksctl version
0.224.0
root@ubuntu:~# curl -LO https://dl.k8s.io/release/v1.30.0/bin/linux/amd64/kubectl

chmod +x kubectl
sudo mv kubectl /usr/local/bin/
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 49.0M  100 49.0M    0     0   666k      0  0:01:15  0:01:15 --:--:--  646k
root@ubuntu:~# kubectl version --client
Client Version: v1.30.0
Kustomize Version: v5.0.4-0.20230601165947-6ce0bf390ce3
root@ubuntu:~#
```

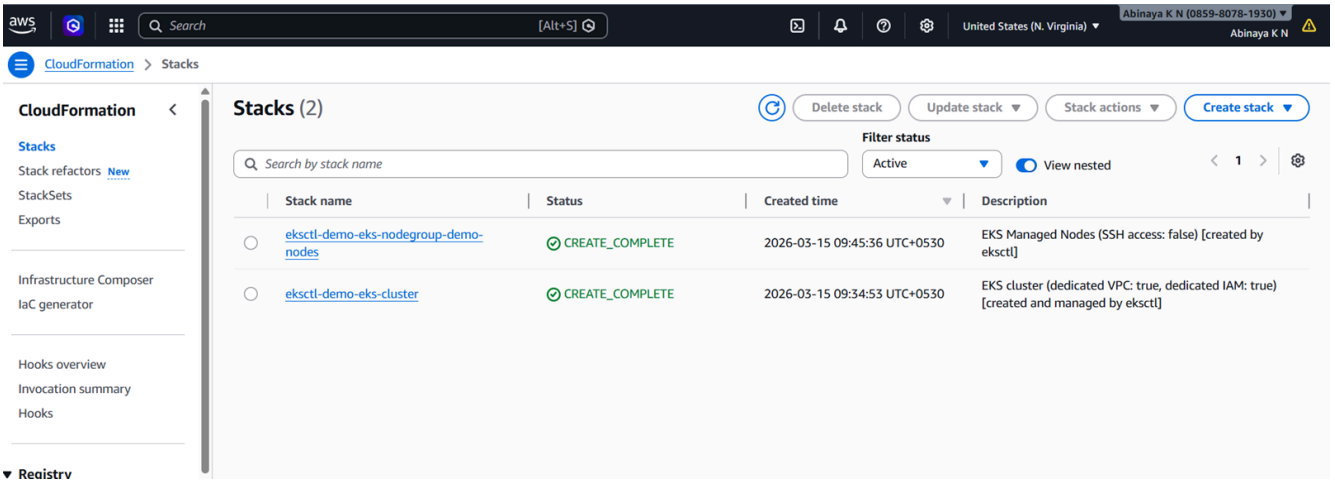
AWS configured

```
abi@ubuntu:~$ aws configure
AWS Access Key ID [None]: AKIARIBG2[redacted]ZKFBH6
AWS Secret Access Key [None]: 34E9k/8jf5MhUKNl[redacted]sEpYetoH9y2g3b3
Default region name [None]: us-west-2
Default output format [None]: json
abi@ubuntu:~$ aws sts get-caller-identity
{
    "UserId": "AIDARIBG2TVV[redacted]LZG5",
    "Account": "085[redacted]31930",
    "Arn": "arn:aws:iam::085[redacted]):user/jenkins"
}
abi@ubuntu:~$
```

EKS Cluster created

```
2026-03-15 03:58:30 [✔]  all cluster resources were deleted
abi@ubuntu:~$ eksctl create cluster \
--name demo-eks \
--region us-east-1 \
--nodegroup-name demo-nodes \
--node-type t3.small \
--nodes 1
2026-03-15 04:04:48 [ℹ]  eksctl version 0.224.0
2026-03-15 04:04:48 [ℹ]  using region us-east-1
2026-03-15 04:04:51 [ℹ]  setting availability zones to [us-east-1f us-east-1a]
2026-03-15 04:04:51 [ℹ]  subnets for us-east-1f - public:192.168.0.0/19 private:192.168.64.0/19
2026-03-15 04:04:51 [ℹ]  subnets for us-east-1a - public:192.168.32.0/19 private:192.168.96.0/19
2026-03-15 04:04:51 [ℹ]  nodegroup "demo-nodes" will use "" [AmazonLinux2023/1.34]
2026-03-15 04:04:51 [!]  Auto Mode will be enabled by default in an upcoming release of eksctl. This means managed node groups and managed networking add-ons
will no longer be created by default. To maintain current behavior, explicitly set 'autoModeConfig.enabled: false' in your cluster configuration. Learn more:
https://eksctl.io/usage/auto-mode/
2026-03-15 04:04:51 [ℹ]  using Kubernetes version 1.34
2026-03-15 04:04:51 [ℹ]  creating EKS cluster "demo-eks" in "us-east-1" region with managed nodes
2026-03-15 04:04:51 [ℹ]  will create 2 separate CloudFormation stacks for cluster itself and the initial managed nodegroup
2026-03-15 04:04:51 [ℹ]  if you encounter any issues, check CloudFormation console or try 'eksctl utils describe-stacks --region=us-east-1 --cluster=demo-eks'
2026-03-15 04:04:51 [ℹ]  Kubernetes API endpoint access will use default of {publicAccess=true, privateAccess=false} for cluster "demo-eks" in "us-east-1"
2026-03-15 04:04:51 [ℹ]  CloudWatch logging will not be enabled for cluster "demo-eks" in "us-east-1"
2026-03-15 04:04:51 [ℹ]  you can enable it with 'eksctl utils update-cluster-logging --enable-types={SPECIFY-YOUR-LOG-TYPES-HERE (e.g. all)} --region=us-east-
1 --cluster=demo-eks'
2026-03-15 04:04:51 [ℹ]  default addons vpc-cni, kube-proxy, coredns, metrics-server were not specified, will install them as EKS addons
2026-03-15 04:04:51 [ℹ]
2 sequential tasks: { create cluster control plane "demo-eks",
    2 sequential sub-tasks: {
        2 sequential sub-tasks: {
            1 task: { create addons },
            wait for control plane to become ready,
        },
        create managed nodegroup "demo-nodes",
    }
}
2026-03-15 04:04:51 [ℹ]  building cluster stack "eksctl-demo-eks-cluster"
2026-03-15 04:04:53 [ℹ]  deploying stack "eksctl-demo-eks-cluster"
2026-03-15 04:05:23 [ℹ]  waiting for CloudFormation stack "eksctl-demo-eks-cluster"
2026-03-15 04:05:56 [ℹ]  waiting for CloudFormation stack "eksctl-demo-eks-cluster"
2026-03-15 04:06:58 [ℹ]  waiting for CloudFormation stack "eksctl-demo-eks-cluster"
```

```
2026-03-15 04:06:58 [ℹ]  waiting for CloudFormation stack "eksctl-demo-eks-cluster"
2026-03-15 04:08:00 [ℹ]  waiting for CloudFormation stack "eksctl-demo-eks-cluster"
2026-03-15 04:09:02 [ℹ]  waiting for CloudFormation stack "eksctl-demo-eks-cluster"
2026-03-15 04:10:04 [ℹ]  waiting for CloudFormation stack "eksctl-demo-eks-cluster"
2026-03-15 04:11:07 [ℹ]  waiting for CloudFormation stack "eksctl-demo-eks-cluster"
2026-03-15 04:12:09 [ℹ]  waiting for CloudFormation stack "eksctl-demo-eks-cluster"
2026-03-15 04:13:11 [ℹ]  waiting for CloudFormation stack "eksctl-demo-eks-cluster"
2026-03-15 04:13:21 [!]  recommended policies were found for "vpc-cni" addon, but since OIDC is disabled on the cluster, eksctl cannot configure the requested
 permissions; the recommended way to provide IAM permissions for "vpc-cni" addon is via pod identity associations; after addon creation is completed, add all
recommended policies to the config file, under `addon.PodIdentityAssociations`, and run `eksctl update addon`
2026-03-15 04:13:21 [ℹ]  creating addon: vpc-cni
2026-03-15 04:13:22 [ℹ]  successfully created addon: vpc-cni
2026-03-15 04:13:23 [ℹ]  creating addon: kube-proxy
2026-03-15 04:13:24 [ℹ]  successfully created addon: kube-proxy
2026-03-15 04:13:24 [ℹ]  creating addon: coredns
2026-03-15 04:13:25 [ℹ]  successfully created addon: coredns
2026-03-15 04:15:34 [ℹ]  building managed nodegroup stack "eksctl-demo-eks-nodegroup-demo-nodes"
2026-03-15 04:15:37 [ℹ]  deploying stack "eksctl-demo-eks-nodegroup-demo-nodes"
2026-03-15 04:15:37 [ℹ]  waiting for CloudFormation stack "eksctl-demo-eks-nodegroup-demo-nodes"
2026-03-15 04:16:10 [ℹ]  waiting for CloudFormation stack "eksctl-demo-eks-nodegroup-demo-nodes"
2026-03-15 04:17:06 [ℹ]  waiting for CloudFormation stack "eksctl-demo-eks-nodegroup-demo-nodes"
2026-03-15 04:19:00 [ℹ]  waiting for CloudFormation stack "eksctl-demo-eks-nodegroup-demo-nodes"
2026-03-15 04:19:01 [ℹ]  waiting for the control plane to become ready
2026-03-15 04:19:05 [✔]  saved kubeconfig as "/home/abi/.kube/config"
2026-03-15 04:19:05 [ℹ]  no tasks
2026-03-15 04:19:05 [✔]  all EKS cluster resources for "demo-eks" have been created
2026-03-15 04:19:08 [ℹ]  nodegroup "demo-nodes" has 1 node(s)
2026-03-15 04:19:08 [ℹ]  node "ip-192-168-49-20.ec2.internal" is ready
2026-03-15 04:19:08 [ℹ]  waiting for at least 1 node(s) to become ready in "demo-nodes"
2026-03-15 04:19:09 [ℹ]  nodegroup "demo-nodes" has 1 node(s)
2026-03-15 04:19:09 [ℹ]  node "ip-192-168-49-20.ec2.internal" is ready
2026-03-15 04:19:09 [✔]  created 1 managed nodegroup(s) in cluster "demo-eks"
2026-03-15 04:19:11 [ℹ]  creating addon: metrics-server
2026-03-15 04:19:12 [ℹ]  successfully created addon: metrics-server
2026-03-15 04:19:19 [ℹ]  kubectl command should work with "/home/abi/.kube/config", try 'kubectl get nodes'
2026-03-15 04:19:19 [✔]  EKS cluster "demo-eks" in "us-east-1" region is ready
abi@ubuntu:~$ kubectl get nodes
NAME                                           STATUS   ROLES    AGE     VERSION
ip-192-168-49-20.ec2.internal   Ready    <none>   3m47s   v1.34.4-eks-f69f56f
abi@ubuntu:~$
```



Ngnix Deployed and Exposed outside the cluster

```
abi@ubuntu:~$ kubectl create deployment nginx --image=nginx
deployment.apps/nginx created
abi@ubuntu:~$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-66686b6766-4mrtf   1/1     Running   0          26s
abi@ubuntu:~$
```

ubuntu | 5% | 0.67 GB / 3.78 GB | 0.01 Mb/s | 0.00 Mb/s | 16 hours | abi (x2) | /: 49% | /boot: 13%

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```
abi@ubuntu:~$ kubectl get nodes
NAME                                           STATUS   ROLES    AGE     VERSION
ip-192-168-49-20.ec2.internal   Ready    <none>   3m47s   v1.34.4-eks-f69f56f
abi@ubuntu:~$ kubectl create deployment nginx --image=nginx
deployment.apps/nginx created
abi@ubuntu:~$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-66686b6766-4mrtf   1/1     Running   0          26s
abi@ubuntu:~$ kubectl expose deployment nginx --type=LoadBalancer --port=80
service/nginx exposed
abi@ubuntu:~$ kubectl get svc
NAME         TYPE           CLUSTER-IP      EXTERNAL-IP                                                              PORT(S)        AGE
kubernetes   ClusterIP      10.100.0.1      <none>                                                                   443/TCP        13m
nginx        LoadBalancer   10.100.105.68   aad53c18a117f4c7d93636854f25003f-462894922.us-east-1.elb.amazonaws.com   80:32114/TCP   18s
abi@ubuntu:~$
```

ubuntu | 5% | 0.65 GB / 3.78 GB | 0.01 Mb/s | 0.00 Mb/s | 16 hours | abi (x2) | /: 49% | /boot: 13%

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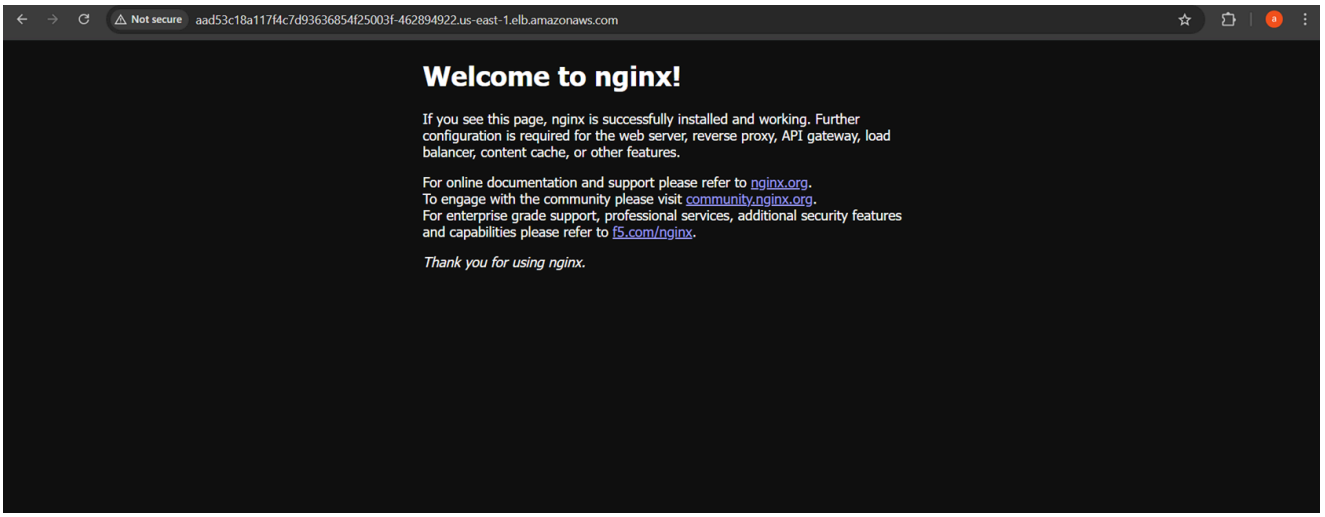
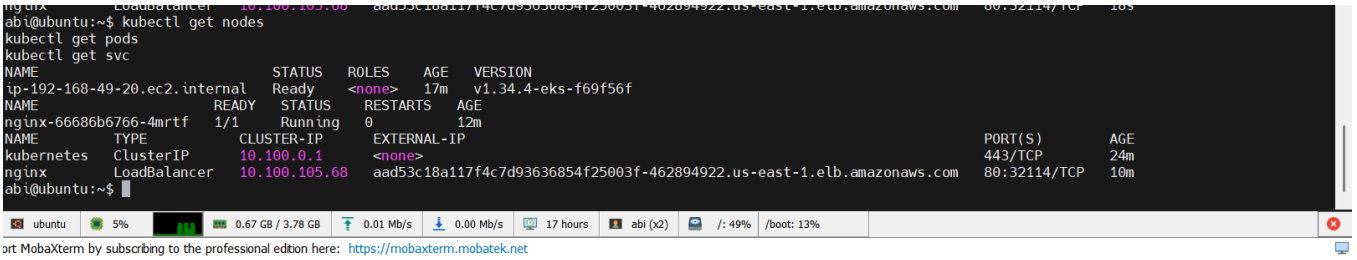

Not secure
aad53c18a117f4c7d93636854f25003f-462894922.us-east-1.elb.amazonaws.com
Welcome to nginx!
If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.
For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.
Thank you for using nginx.


abi@ubuntu:~$ kubectl get nodes
kubectl get pods
kubectl get svc
NAME STATUS ROLES AGE VERSION
ip-192-168-49-20.ec2.internal Ready <none> 17m v1.34.4-eks-f69f56f
NAME READY STATUS RESTARTS AGE
nginx-66686b6766-4mrtf 1/1 Running 0 12m
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
kubernetes ClusterIP 10.100.0.1 <none> 443/TCP 24m
nginx LoadBalancer 10.100.105.68 aad53c18a117f4c7d93636854f25003f-462894922.us-east-1.elb.amazonaws.com 80:32114/TCP 10m
abi@ubuntu:~$
ubuntu
5%
0.67 GB / 3.78 GB
0.01 Mb/s
0.00 Mb/s
17 hours
abi (x2)
/: 49%
/boot: 13%
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